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Inventor(s)	Kim; Changhee et al.

Semiconductor integrated circuit device and manufacturing method thereof

Abstract

A semiconductor integrated circuit device including a substrate with a first element region of a P type and a second element region of an N type, a channel active region that extends in the first element region or the second element region, the channel active region including a plurality of channels, a plurality of gate lines that extend in a second direction intersecting and include a gate metal layer, and a gate insulating film in contact with the gate metal layer, a plurality of first spacers on opposite side portions of respective ones of the gate lines, and a plurality of source/drain regions that are between ones of the plurality of gate lines. The channel active region includes a first channel directly on the substrate, and a second channel spaced apart from the first channel and extends into the gate metal layer.

Inventors: **Kim; Changhee (Suwon-si, KR), Kim; Kyungsoo (Hwaseong-si, KR), Bae; Dongil (Seongnam-si, KR), Whang; Sungman (Yongin-si, KR)**

Applicant: **SAMSUNG ELECTRONICS CO., LTD.** (Suwon-si, KR)

Family ID: **1000008779625**

Assignee: **SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)**

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8759168	12/2013	Cheng et al.	N/A	N/A
9305846	12/2015	Jacob et al.	N/A	N/A
9306001	12/2015	Cai et al.	N/A	N/A
9590038	12/2016	Kim et al.	N/A	N/A
9716142	12/2016	Bi et al.	N/A	N/A
10431585	12/2018	Yang et al.	N/A	N/A
10431651	12/2018	Chao et al.	N/A	N/A
10431690	12/2018	Rachmady et al.	N/A	N/A
10580897	12/2019	Pawlak	N/A	N/A
2013/0302962	12/2012	Cheng et al.	N/A	N/A
2015/0140761	12/2014	Jacob et al.	N/A	N/A
2017/0104062	12/2016	Bi et al.	N/A	N/A
2018/0158957	12/2017	Rachmady et al.	N/A	N/A
2018/0175035	12/2017	Yang et al.	N/A	N/A
2018/0374946	12/2017	Pawlak	N/A	N/A
2021/0066506	12/2020	Liaw	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
109755290	12/2018	CN	N/A
20170048112	12/2016	KR	N/A

Primary Examiner: Kim; Jay C

Assistant Examiner: Lee; Woo K

Attorney, Agent or Firm: MORGAN, LEWIS & BOCKIUS LLP

Background/Summary

CROSS-REFERENCE TO THE RELATED APPLICATION

(1) This application claims priority from Korean Patent Application No. 10-2021-0075926, filed on Jun. 11, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

(2) The example embodiments of the disclosure relate to a semiconductor integrated circuit device and a manufacturing method thereof.

(3) In accordance with rapid advance of electronic industries and user demand, electronic appliances are being further miniaturized and multifunctionalized. Accordingly, down-scaling of a semiconductor integrated circuit device is rapidly progressing, and the line width and pitch of a multilayer wiring structure included in the semiconductor integrated circuit are further decreasing.

(4) As a result, there may be a phenomenon in which current leakage through a channel between adjacent source/drain regions occurs. In order to reduce such a current leakage phenomenon, a structure has been proposed in which an insulating (dielectric) material is formed between a source/drain region and a substrate in order to space the source/drain region and the substrate apart from each other.

SUMMARY

(5) The example embodiments of the disclosure provide a semiconductor integrated circuit device having a structure capable of minimizing or reducing current leakage between adjacent source/drain regions without formation of an insulating dielectric material spacing a source/drain region and a substrate apart from each other, and a manufacturing method thereof.

(6) According to some embodiments of the present disclosure, there is provided a semiconductor integrated circuit device. The semiconductor integrated circuit device includes a substrate including a first element region of a P type and a second element region of an N type, a channel active region that extends into the first element region or the second element region in a first direction, the channel active region including a plurality of channels, a plurality of gate lines that extend in a second direction intersecting the first direction and include a gate metal layer, and a gate insulating film in contact with the gate metal layer, a plurality of first spacers on opposite side portions of the plurality of gate lines and separated from one another, and a plurality of source/drain regions that are between the plurality of gate lines. The channel active region includes a first channel directly on the substrate, and a second channel spaced apart from the first channel in a third direction perpendicular to the first direction and the second direction and extends into the gate metal layer. A width in the first direction of the second channel is greater than respective widths in the first direction of the gate metal layer and the gate insulating film that are between the first channel and the second channel. The first channel, the second channel, and portions of the gate lines between the first channel and the second channel have a first groove therebetween. A first spacer of the first spacers includes a first section in the first groove and a second section that penetrates a first source/drain region of the plurality of source/drain regions.

(7) According to some embodiments of the present disclosure, there is provided a semiconductor integrated circuit device. The semiconductor integrated circuit device includes a substrate including a first element region of a P type and a second element region of an N type, a channel active region that extends into the first element region or the second element region in a first direction, the

channel active region includes a plurality of channels, a plurality of gate lines that extend in a second direction intersecting the first direction and each of the plurality of gate lines includes a gate metal layer and a gate insulating film in contact with the gate metal layer, source/drain regions that are between the plurality of gate lines such that a first source/drain region includes a first epitaxial region and a second epitaxial region of different conductivity types, and a plurality of first spacers on opposite side portions of the plurality of gate lines. A first spacer of the plurality of first spacers is between the first epitaxial region and the second epitaxial region of the first source/drain region, and overlaps a respective one of the plurality of gate lines in the first direction.

(8) According to some embodiments of the present disclosure, there is provided a semiconductor integrated circuit device. The semiconductor integrated circuit device includes a substrate including a first element region of a P type and a second element region of an N type, a channel active region that extends into the first element region or the second element region in a first direction, the channel active region including a plurality of channels, a plurality of gate lines that extend in a second direction intersecting the first direction and each includes a gate metal layer and a gate insulating film in contact with the gate metal layer, a plurality of spacers on opposite side portions of the gate lines and are separated from one another, and a plurality of source/drain regions that are between the plurality of gate lines. The channel active region includes a first channel directly on the substrate, a second channel spaced apart from the first channel in a third direction perpendicular to the first direction and the second direction and extending into the gate metal layer, and a third channel spaced apart from the second channel in the third direction and extends into the gate metal layer. Respective widths in the first direction of the second channel and the third channel are greater than respective widths in the first direction of the gate metal layer and the gate insulating film that are between the first channel and the second channel. The first channel, the second channel, and respective portions of first ones of the gate lines between the first channel and the second channel have a second groove therebetween. The second channel, the third channel, respective portions of ones of the gate lines between the second channel and the third channel form a second groove. The plurality of spacers includes a first spacer including a first section in the first groove, and a second section that penetrates a first source/drain region of the plurality of source/drain regions, a second spacer in the second groove, a third spacer on the third channel at an outside of a part of a respective one of the gate lines, and a fourth spacer on the third channel at an outside of the third spacer. The first source/drain region includes a first epitaxial region and a second epitaxial region of different conductivity types. The first spacer is directly on the first epitaxial region and contacts the second epitaxial region. The second spacer is directly on the second epitaxial region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic layout view of a semiconductor integrated circuit device according to example embodiments of the disclosure.

(2) FIG. **2** is a schematic cross-sectional view of the semiconductor integrated circuit device taken along line I-I′ in FIG. **1**.

(3) FIG. **3** is an enlarged view of a portion A of FIG. **2**.

(4) FIG. **4** is a schematic cross-sectional view of the semiconductor integrated circuit device taken along line II-II′ in FIG. **1**.

(5) FIGS. **5** to **12** are views schematically showing procedures of a method for manufacturing a semiconductor integrated circuit device in accordance with example embodiments of the disclosure, respectively.

(6) FIG. **13** is a schematic cross-sectional view of a semiconductor integrated circuit device

according to example embodiments of the disclosure.
(7) FIGS. **14** to **23** are views schematically showing procedures of a method for manufacturing the semiconductor integrated circuit device of FIG. **13**.
(8) FIG. **24** is a schematic cross-sectional view of a semiconductor integrated circuit device according to example embodiments of the disclosure.
(9) FIGS. **25** to **32** are views schematically showing procedures of a method for manufacturing the semiconductor integrated circuit device of FIG. **24**.
(10) FIG. **33** is a schematic cross-sectional view of a semiconductor integrated circuit device according to example embodiments of the disclosure.
(11) FIG. **34** is a schematic cross-sectional view of a semiconductor integrated circuit device according to example embodiments of the disclosure.
(12) FIG. **35** is a schematic cross-sectional view of a semiconductor integrated circuit device according to example embodiments of the disclosure.
(13) FIG. **36** is a schematic cross-sectional view of a semiconductor integrated circuit device according to an example embodiment of the disclosure.
(14) FIG. **37** is a schematic cross-sectional view of a semiconductor integrated circuit device according to example embodiments of the disclosure.
(15) FIG. **38** is a schematic cross-sectional view of a semiconductor integrated circuit device according to example embodiments of the disclosure.

DETAILED DESCRIPTION

(16) FIG. **1** is a schematic layout view of a semiconductor integrated circuit device according to example embodiments of the disclosure.
(17) In the specification, a first direction D**1** and a second direction D**2** may refer to horizontal directions, that is, directions intersecting on a horizontal plane, respectively, and a third direction D**3** may refer to a vertical direction. For example, the first direction D**1** may correspond to an X-axis direction, the second direction D**2** may correspond to a Y-axis direction, and the third direction D**3** may correspond to a Z-axis direction. The plane formed in the first direction D**1** and the second direction D**2** may be a horizontal plane.
(18) Referring to FIG. **1**, the semiconductor integrated circuit device may include a first-type well having a predetermined width and extending in the first direction D**1**, and a second-type well disposed adjacent to the first-type well while having a predetermined width and extending in the first direction D**1**. Here, a region where the first-type well is formed is defined as a first element region RX**1**, and a region where the second-type well is formed is defined as a second element region RX**2**. In some embodiments, the first-type well is a well of one of a P type or an N type, and the second-type well is a well of the other of the P type or the N type.
(19) Although not clearly shown, the first element region RX**1** and the second element region RX**2** may be adjacent to each other in the second direction D**2**. In some embodiments, the first-type well and the second-type well may be formed on a substrate (cf. "**101**" in FIG. **2**). That is, the first element region RX**1** and the second element region RX**2** may be regions on the substrate. In some embodiments, widths of the first element region RX**1** and the second element region RX**2** (widths in the second direction D**2**) are equal.
(20) The substrate may include a semiconductor such as Si or Ge, or a compound semiconductor such as SiGe, SiC, GaAs, InAs, or InP. The substrate may include a conductive region, for example, a well doped with impurities or a structure doped with impurities.
(21) In some embodiments, the first element region RX**1** and the second element region RX**2**, which are formed with a plurality of channel active regions **111** and **112** (or fin-type channel active regions), and an active cut region ACR separating the first element region RX**1** and the second element region RX**2** from each other may be defined at the substrate. Each of the first element region RX**1**, the active cut region ACR, and the second element region RX**2** may extend in the first direction D**1**. The first element region RX**1**, the active cut region ACR, and the second element

region RX**2** may be arranged in the second direction D**2**.

(22) The plurality of channel active regions **111** and **112** may have a shape protruding from the substrate or may be formed on the substrate. The plurality of channel active regions **111** and **112** may extend in parallel in the first direction D**1**. The plurality of channel active regions may be arranged in the second direction D**2**. The plurality of channel active regions may include a first channel active region **111** formed in the first element region RX**1**, and a second channel active region **112** formed in the second element region RX**2**. In some embodiments, the first channel active region **111** is of the P type, and the second channel active region **112** may be of the N type.

(23) The semiconductor integrated circuit device may include a plurality of gate lines **220** extending in the second direction D**2**. Each gate line **220** may intersect the plurality of channel active regions **111** and **112**. The semiconductor integrated circuit device may further include a transistor and additional patterns for routing in accordance with a desired function, on the basis of the structure of the semiconductor integrated circuit device. In some embodiments, a part of the gate lines **220** may have a shape in which the gate lines **220** are divided in the second direction D**2** by, for example, an etching process. For example, a part of the gate lines **220** extending in the second direction D**2** may be divided by the gate cut region CT.

(24) The gate line **220** may include a layer containing a work function metal and a gap-fill metal film. For example, the layer containing the work function metal may include at least one metal of Ti, W, Ru, Nb, Mo, Hf, Ni, Co, Pt, Yb, Tb, Dy, Er or Pd, and the gap-fill metal film may be a W film or an Al film. In some embodiments, the gate lines **220** may include a stack structure of TiAlC/TiN/W, a stack structure of TiN/TaN/TiAlC/TiN/W, or a stack structure of TiN/TaN/TiN/TiAlC/TiN/W.

(25) The semiconductor integrated circuit device may include a first power line **301** extending in the first direction D**1** while being configured to receive a first voltage, and a second power line **302** extending in the first direction D**1** while being configured to receive a second voltage lower than the first voltage. The first power line **301** and the second power line **302** may be disposed at one wiring layer M**1** of a back-end-of-line (BEOL) structure on a front-end-of-line (FEOL) structure.

(26) In accordance with some embodiments, the first voltage may be a positive voltage, and the second voltage may be a negative voltage or a ground voltage. The first power line **301** and the second power line **302** may be spaced apart from each other by a predetermined pitch, and may be alternately disposed in the second direction D**2**. For example, the first power line **301** may be disposed on the first element region RX**1**, and the second power line **302** may be disposed on the second element region RX**2**. In accordance with some embodiments, the first power line **301** and the second power line **302** may be formed to overlap with portions of the gate lines **220**. The first power line **301** and the second power line **302** may be electrically connected to the gate lines **220** and/or the source/drain regions **230**.

(27) The source/drain regions **230** may be formed at opposite sides of one gate line **220**. The source/drain regions **230** may be formed among the gate lines **220**. The source/drain regions **230** may overlap with the first channel active region **111** and the second channel active region **112**. In some embodiments, the source/drain region **230** may be configured by an impurity ion implantation region formed at portions of the first channel active region **111** and the second channel active region **112**, a semiconductor epitaxial layer epitaxially grown from recess regions formed in the channel active regions **111** and **112**, or a combination thereof.

(28) A plurality of gate cut regions CT may be formed to overlap with the first power line **301**, the second power line **302** and the active cut region ACR. A part of the gate cut regions CT may have a shape extending in the first direction D**1** while overlapping with the first power line **301** and the second power line **302**. Another part of the gate cut regions CT may be formed to overlap with the active cut region ACR, thereby dividing a part of the gate lines **220** extending in the second direction D**2**.

(29) FIG. **2** is a schematic cross-sectional view of the semiconductor integrated circuit device taken

along line I-I′ in FIG. **1**. FIG. **3** is an enlarged view of a portion A of FIG. **2**. FIG. **4** is a schematic cross-sectional view of the semiconductor integrated circuit device taken along line II-IF in FIG. **1**.
(30) Referring to FIGS. **1** to **4**, a substrate **101** may include the channel active regions **111** and **112** which have a shape protruding upwards from one surface (for example, an active surface) of the substrate **101**. In FIGS. **2** to **4**, only the first channel region **111**, which is formed in the first element region RX**1**, is illustrated.
(31) An FEOL structure may be disposed on the substrate **101**. The FEOL structure may be formed by an FEOL process. The FEOL process may refer to a process for forming individual elements, for example, a transistor, a capacitor, a resistor, etc., on the substrate **101** in a manufacturing procedure for the integrated circuit chip. For example, the FEOL process may include planarization and cleaning of a wafer, formation of a trench, formation of a well, formation of a gate line, formation of a source and drain, etc.
(32) In some embodiments, the FEOL structure may include a logic cell including a multi-bridge channel FET (MBCFET). Of course, the FEOL structure is not limited to the above-described condition, and may include a logic cell including a metal-oxide-semiconductor field effect transistor (MOSFET), a fin field effect transistor (FinFET), a system large scale integration (LSI) device, a microelectromechanical system (MEMS), an active device, or a passive device which includes a plurality of transistors, so long as the scope and spirit of the disclosure are not changed. In the following description, description of the example embodiments of the disclosure will be given mainly in conjunction with the substrate and the FEOL structure.
(33) The semiconductor integrated circuit device may include a lower insulating film **151** disposed on the substrate **101**. The lower insulating film **151** may extend to have a predetermined thickness on opposite side portions of the channel active regions **111** and **112** and on the substrate **101**. The lower insulating film **151** may be disposed around regions beneath opposite side surfaces of the channel active regions **111** and **112**. The lower insulating film **151** may include an insulating material. For example, the lower insulating film **151** may include silicon oxide, silicon nitride and/or silicon oxynitride.
(34) Each of the channel active regions **111** and **112** may include a plurality of channels formed to be spaced apart from one another. For example, each of the channel active regions **111** and **112** may include first to fourth channels CH**1** to CH**4**. The first to fourth channels CH**1** to CH**4** may include the same material. The function and material of each of the channels CH**1** to CH**4** may be varied in accordance with whether the semiconductor integrated circuit device is of a PMOS type or an NMOS type.
(35) The first channel CH**1** may be a channel protruding from the substrate **101**. The first channel CH**1** may be a region of an upper portion of the substrate **101** and a region directly protruding from the substrate **101** in the third direction D**3**. For example, one first channel CH**1** may be provided for each channel active region **111**. In accordance with some embodiments, the first channel CH**1** may not substantially perform a channel function of a transistor.
(36) The second channel CH**2** may be spaced apart from the first channel CH**1** in the third direction D**3**. The second channel CH**2** may extend through the gate line **220** while extending in the first direction D**1**. The second channel CH**2** may have a wire pattern shape. The second channel CH**2** may overlap with the first channel CH**1** in the third direction D**3**. In some embodiments, the width of the second channel CH**2** may be equal to the width of the first channel CH**1**. The second channel CH**2** may perform a channel function of a transistor.
(37) The third channel CH**3** may be spaced apart from the second channel CH**2** in the third direction D**3**. The third channel CH**3** may extend through the gate line **220** while extending in the first direction D**1**. The third channel CH**3** may have a wire pattern shape. The third channel CH**3** may overlap with the first channel CH**1** and the second channel CH**2** in the third direction D**3**. In some embodiments, the width of the third channel CH**3** may be equal to the width of the first channel CH**1**. The third channel CH**3** may perform a channel function of a transistor.

(38) The fourth channel CH**4** may be spaced apart from the third channel CH**3** in the third direction D**3**. The fourth channel CH**4** may extend through the gate line **220** while extending in the first direction D**1**. The fourth channel CH**4** may have a wire pattern shape. The fourth channel CH**4** may overlap with the first to third channels CH**1** to CH**3** in the third direction D**3**. In some embodiments, the width of the fourth channel CH**4** may be equal to the width of the first channel CH**1**. The fourth channel CH**4** may perform a channel function of a transistor.
(39) In some embodiments, each width of the first to fourth channels CH**1** to CH**4** may be greater than the width (for example, the width in the first direction D**1**) of the gate line **220**. The first channel CH**1** and the second channel CH**2** may include a first groove GV**1** together with a portion of the gate line **220** formed between the first channel CH**1** and the second channel CH**2**. The second channel CH**2** and the third channel CH**3** may include a second groove GV**2** together with a portion of the gate line **220** formed between the second channel CH**2** and the third channel CH**3**. The third channel CH**3** and the fourth channel CH**4** may include a third groove GV**3** together with a portion of the gate line **220** formed between the third channel CH**3** and the fourth channel CH**4**. The first to third grooves GV**1** to GV**3** may be formed such that two grooves are disposed at opposite sides with reference to one gate line **220**, respectively.
(40) In accordance with some embodiments, at least one of the third channel CH**3** and the fourth channel CH**4** may be omitted.
(41) A gate insulating film **152** may be disposed on a portion of the lower insulating film **151**. In addition, the gate insulating film **152** may be disposed on the channel active regions **111** and **112**. For example, the gate insulating film **152** may be formed between each of the first to fourth channels CH**1** to CH**4** and a gate metal layer **154**. The gate insulating film **152** may be conformally formed along peripheries of the second to fourth channels CH**2** to CH**4**.
(42) The gate insulating film **152** may include a silicon oxide film, a high-k dielectric film, or a combination thereof. The gate insulating film **152** may be formed by an atomic layer deposition (ALD) process, a chemical vapor deposition (CVD) process or a physical vapor deposition (PVD) process.
(43) In accordance with some embodiments, the semiconductor integrated circuit device may include a negative capacitance (NC) FET using a negative capacitor. For example, the gate insulating film **152** may include a ferroelectric material film having ferroelectric characteristics, and a paraelectric material film having paraelectric characteristics.
(44) The ferroelectric material film may have a negative capacitance, and the paraelectric material film may have a positive capacitance. For example, when two or more capacitors are connected in series, and the capacitance of each of the capacitors has a positive value, the total capacitance of the capacitors may be lower than the capacitance of each individual capacitor. On the other hand, when at least one of the capacitances of two or more capacitors connected in series has a negative value, the total capacitance of the capacitors may have a positive value and may be greater than an absolute value of the capacitance of each individual capacitor.
(45) When a ferroelectric material film having a negative capacitance and a paraelectric material film having a positive capacitance are connected in series, the total capacitance of the ferroelectric material film and the paraelectric material film connected in series may increase. A transistor including a ferroelectric material film may have subthreshold swing (SS) of less than 60 mV/decade at normal temperature, using an increase in total capacitance as described above.
(46) The ferroelectric material film may have ferroelectric dielectric characteristics. The ferroelectric material film may include, for example, at least one of hafnium oxide, hafnium zirconium oxide, barium strontium titanium oxide, barium titanium oxide, and/or lead zirconium titanium oxide. Here, for example, hafnium zirconium oxide may be a material produced by doping hafnium oxide with zirconium (Zr). In another example, hafnium zirconium oxide may be a compound of hafnium (Hf), zirconium (Zr), and oxygen (O).
(47) The ferroelectric material film may further include a dopant doped therein. For example, the

dopant may include at least one of aluminum (Al), titanium (Ti), niobium (Nb), lanthanum (La), yttrium (Y), magnesium (Mg), silicon (Si), calcium (Ca), cerium (Ce), dysprosium (Dy), erbium (Er), gadolinium (Gd), germanium (Ge), scandium (Sc), strontium (Sr), and/or tin (Sn). The kind of the dopant included in the ferroelectric material film may be varied in accordance with which ferroelectric material is included in the ferroelectric material film.
(48) When the ferroelectric material film includes hafnium oxide, the dopant included in the ferroelectric material film may include, for example, at least one of gadolinium (Gd), silicon (Si), zirconium (Zr), aluminum (Al), and/or yttrium (Y).
(49) When the dopant is aluminum (Al), the ferroelectric material film may include 3 to 8 atomic % (at %) of aluminum. Here, the ratio of the dopant may be the ratio of aluminum to the sum of hafnium and aluminum.
(50) When the dopant is silicon (Si), the ferroelectric material film may include 2 to 10 at % of silicon. When the dopant is yttrium (Y), the ferroelectric material film may include 2 to 10 at % of yttrium. When the dopant is gadolinium (Gd), the ferroelectric material film may include 1 to 7 at % of gadolinium. When the dopant is zirconium (Zr), the ferroelectric material film may include 50 to 80 at % of zirconium.
(51) The paraelectric material film may have paraelectric characteristics. The paraelectric material film may include, for example, at least one of silicon oxide and a metal oxide having a high dielectric constant. The metal oxide included in the paraelectric material film may include, for example, at least one of hafnium oxide, zirconium oxide, and aluminum oxide, without being limited thereto.
(52) The ferroelectric material film and the paraelectric material film may include the same material. The ferroelectric material film has ferroelectric characteristics, but the paraelectric material film may not have ferroelectric characteristics. For example, when both the ferroelectric material film and the paraelectric material film include hafnium oxide, the crystalline structure of the hafnium oxide included in the ferroelectric material film may differ from the crystalline structure of the hafnium oxide included in the paraelectric material film.
(53) The ferroelectric material film may have a thickness exhibiting ferroelectric characteristics. The thickness of the ferroelectric material film may be, for example, 0.5 to 10 nm, without being limited thereto. The critical thickness exhibiting ferroelectric characteristics may be varied in accordance with different ferroelectric materials and, as such, the thickness of the ferroelectric material film may be varied in accordance with the ferroelectric material thereof.
(54) For example, the gate insulating film **152** may include one ferroelectric material film. In another example, the gate insulating film **152** may include a plurality of ferroelectric material films spaced apart from one another. In this case, the gate insulating film **152** may have a stacked film structure in which a plurality of ferroelectric material films and a plurality of paraelectric material films are alternately stacked.
(55) The gate metal layer **154** may be disposed on the gate insulating film **152** and among the first to fourth channels CH**1** to CH**4**. The gate metal layer **154** may function to adjust a work function. For example, the gate metal layer **154** may include at least one of TiN, WN, TaN, Ru, TiC, TaC, Ti, Ag, Al, TiAl, TiAlN, TiAlC, TaCN, TaSiN, Mn, and/or Zr. The gate metal layer **154** and the gate insulating film **152** may form the above-described gate line **220**.
(56) In some embodiments, each source/drain region **230** may include a plurality of sequentially stacked epitaxial regions. For example, each source/drain region **230** may include a first epitaxial region EP**1**, and a second epitaxial region EP**2** disposed on the first epitaxial region EP**1**.
(57) The first epitaxial region EP**1** may be directly disposed on the channel active regions **111** and **112**. The first epitaxial region EP**1** may contact the first channel CH**1**. The second epitaxial region EP**2** may be directly disposed on the first epitaxial region EP**1**.
(58) The second epitaxial region EP**2** may be disposed between the second to fourth channels CH**2** to CH**4** adjacent to each other. The second epitaxial region EP**2** may contact the adjacent second to

fourth channels CH**2** to CH**4**. In some embodiments, the position of a bottom surface of the second epitaxial region EP**2** may be lower than the position where the second channel CH**2** is formed, and the position of a top surface of the second epitaxial region EP**2** may be higher than the position where the fourth channel CH**4** is formed. In some embodiments, the thickness (the thickness in a vertical direction) of the second epitaxial region EP**2** may be greater than the thickness of the first epitaxial region EP**1**.
(59) In some embodiments, the first epitaxial region EP**1** and the second epitaxial region EP**2** may be of different types, respectively. In some embodiments, the first epitaxial region EP**1** may be of the same type as the element region of the substrate **101** formed with the source/drain region **230**, and the second epitaxial region EP**2** may be of a type different from that of the first epitaxial region EP**1**. For example, when the first element region RX**1** is of the P type, the first epitaxial region EP**1** of the source/drain region **230** formed in the first element region RX**1** may be of the P type, and the second epitaxial region EP**2** may be of the N type. In this case, the first epitaxial region EP**1** of the source/drain region **230** formed in the second element region RX**2** may be of the N type, and the second epitaxial region EP**2** may be of the P type.
(60) In some embodiments, the first epitaxial region EP**1** and the second epitaxial region EP**2** may be of the same type. In some embodiments, the first epitaxial region EP**1** and the second epitaxial region EP**2** may be of a type different from that of the element region of the substrate **101** formed with the source/drain region **230**. For example, when the first element region RX**1** is of the P type, the first epitaxial region EP**1** of the source/drain region **230** formed in the first element region RX**1** may be of the N type, and the second epitaxial region EP**2** may be of the N type. In this case, the first epitaxial region EP**1** of the source/drain region **230** formed in the second element region RX**2** may be of the P type, and the second epitaxial region EP**2** may be of the P type.
(61) A plurality of spacers may be disposed at opposite side portions of the gate line **220** while being separate from one another. That is, a plurality of spacers may be disposed at side portions of the gate insulating film surrounding the gate metal layer. For example, the spacers may include a first spacer **181** formed to penetrate the source/drain region **230** while filling the first groove GV**1** between the first channel CH**1** and CH**2**, a second spacer **182** formed to fill the second groove G**2** between the second channel CH**2** and the third channel CH**3**, a third spacer **183** formed to fill the third groove GV**3** between the third channel CH**3** and the fourth channel CH**4**, a fourth spacer **157** surrounding opposite side portions of the gate line **220** on the fourth channel CH**4**, and a fifth spacer **158** surrounding an outside of the fourth spacer **157**.
(62) In some embodiments, the first spacer **181** may include a first section SP**1** filling the first groove GV**1** between the first channel CH**1** and the second channel CH**2** while overlapping with the first channel CH**1** and the second channel CH**2** in a vertical direction (the third direction D**3**), and a second section SP**2** penetrating the source/drain region **230**. The first section SP**1** may overlap with the first channel CH**1** and the second channel CH**2** in the third direction D**3**. The second section SP**2** may overlap with the source/drain region **230** in the third direction D**3**. Although the boundary between the first section SP**1** and the second section SP**2** is not clearly distinguished, the boundary may be defined to overlap with a line vertically extending along the side surfaces of the second to fourth channels CH**2** to CH**4**.
(63) The first spacer **181** may be formed at each of opposite side portions of the gate line **220** between the first channel CH**1** and the second channel CH**2**. Two first spacers **181** separate from each other may be disposed at opposite side portions of one gate line **220**, respectively. The two first spacers **181** disposed at the opposite side portions of one gate line **220** do not physically contact each other.
(64) The first section SP**1** may be disposed adjacent to an outside of the gate insulating film. The second section SP**2** may be formed between the first epitaxial region EP**1** and the second epitaxial region EP**2**. One side portion of the first section SP**1** may contact the gate insulating film, and the other side portion of the first section SP**1** may contact the second section SP**2**.

(65) The second section SP**2** may be disposed on the second epitaxial region EP**2** adjacent to an edge of the second epitaxial region EP**2**. In some embodiments, one side portion of the second section SP**2** may contact the first section SP**1** and the first channel CH**1**, and the other side portion of the second section SP**2** may contact the first epitaxial region EP**1** and the second epitaxial region EP**2**. In some embodiments, a top surface of the second section SP**2** may contact the first epitaxial region EP**1**, and a bottom surface of the second section SP**2** may contact the second epitaxial region EP**2**. The second section SP**2** may overlap with the first channel CH**1** in the first direction D**1** (horizontal direction). In some embodiments, the thickness of the second section SP**2** may be greater than the thickness of the first section SP**1**.
(66) In some embodiments, the second spacer **182** and the third spacer **183** may not overlap with the source/drain region **230** in the third direction D**3**.
(67) In some embodiments, the first spacer **181**, the second spacer **182**, and the third spacer **183** may include the same material. In some embodiments, the first section SP**1** and the second section SP**2** of the first spacer **181** may include the same material. The first spacer **181**, the second spacer **182**, and the third spacer **183** may include an insulating material. For example, the first spacer **181**, the second spacer **182**, and the third spacer **183** may include SiN, SiON, SiCON, or a combination thereof.
(68) The fourth spacer **157** and the fifth spacer **158** may include an insulating material. The fourth spacer **157** and the fifth spacer **158** may include SiN, SiON, SiCON, or a combination thereof.
(69) A gate capping layer **155** may be disposed on the gate metal layer. Although not clearly shown, a via may be formed at the gate capping layer **155** in order to connect the gate metal layer to a structure thereover. In some embodiments, the gate capping layer **155** may be omitted.
(70) It may be possible to minimize or reduce leakage of current between adjacent ones of the source/drain regions **230** (for example, through the first channel CH**1**) by forming the second epitaxial region EP**2** such that the second epitaxial region EP**2** contacts a portion of the first channel CH**1** disposed at a lowermost end of the channel active regions **111** and **112**, and disposing the second section SP**2** of the first spacer **181** between the first channel CH**1** and the source/drain region **230**.
(71) Hereinafter, a method for manufacturing a semiconductor integrated circuit device will be described. The following description will be given mainly in conjunction with a procedure for manufacturing a first spacer **181** and a procedure for manufacturing a source/drain region **230**.
(72) FIGS. **5** to **12** are views schematically showing procedures of a method for manufacturing a semiconductor integrated circuit device in accordance with example embodiments of the disclosure, respectively.
(73) Referring to operation S**110** of FIG. **5**, a substrate sacrificial material **102***a* and a substrate material **103***a* may be alternately stacked on a base substrate material **101**. The base substrate material **101** and the substrate material **103***a* may include the same material as the above-described substrate **101**. For example, the base substrate material **101** and the substrate material **103***a* may include Si. For example, the substrate sacrificial material **102***a* may include SiGe.
(74) Thereafter, an insulating pattern **161** may be formed on a structure in which the substrate sacrificial material **102***a* and the substrate material **103***a* are alternately stacked, and a plurality of vertical pattern structures, in which a gate sacrificial pattern **162** and a hard mask pattern **163** are stacked, may be formed on the insulating pattern **161**.
(75) Subsequently, an outer spacer material **158***a* covering or overlapping the plurality of vertical pattern structures may be formed. The outer spacer material **158***a* may include the same material as a fifth spacer **158**.
(76) Next, referring to operation S**120** of FIG. **6**, portions, among the plurality of vertical pattern structures, of the base substrate material **101** and the structure, in which the substrate sacrificial material **102***a* and the substrate material **103***a* are alternately stacked, may be removed through a wet etching process. The removed portions may have a predetermined depth. Vertical grooves may

be formed by the removed portions. A side portion of the substrate sacrificial material **102***a* may be exposed at each vertical groove.
(77) A portion of an edge of the substrate sacrificial material **102***a* may be removed in each vertical groove through an over-etching process. A part of the exposed side portion of the substrate sacrificial material **102***a* may be removed. As the portion of the edge of the substrate sacrificial material **102***a* is removed, a horizontal stepped structure may be formed between a substrate sacrificial material **102***b* and a substrate material **103***b*. Two substrate materials **103***b* adjacent to each other and one substrate sacrificial material **102***b* formed therebetween may form a groove in accordance with a stepped structure thereof and, as such, first to third grooves GV**1** to GV**3** may be formed.
(78) Thereafter, referring to operation S**130** of FIG. **7**, a first spacer material **180***a* may be formed to cover or overlap the vertical grooves while filling the first to third grooves GV**1** to GV**3** and to cover or overlap the plurality of vertical pattern structures. The first spacer material **180***a* may include the same material as first to third spacers **181** to **183**.
(79) Subsequently, referring to operation S**140** of FIG. **8**, a coating material **190** may be formed in a region where no process will be performed from among a first element region RX**1** and a second element region RX**2**. For example, the coating material **190** may be formed over the entire surface of the second element region RX**2**, for execution of a process in the first element region RX**1**. For example, the coating material **190** may include SOH.
(80) Thereafter, referring to operation S**150** of FIG. **9**, the coating material **190** may be removed through an etching process, and portions of the first spacer material **180***a* disposed on the hard mask pattern **163** and beneath the vertical grooves may be removed through an etch-back process. The partially-removed first spacer material **180***a* may expose a top surface of the substrate **101** and a top surface of the hard mask pattern **163**. The remaining portion is denoted as first spacer material **180***b*.
(81) Next, referring to operation S**160** of FIG. **10**, a first epitaxial material EP**1***a* and a second epitaxial material EP**2***a* may be sequentially formed on the substrate **101** exposed in the vertical grooves. The first epitaxial material EP**1***a* and the second epitaxial material EP**2***a* may be formed to have predetermined heights extending from the top surface of the substrate **101** while contacting the first spacer material **180***a*. In some embodiments, the height of a top surface of the second epitaxial material EP**2***a* may be equal to or lower than the height of a top surface of the substrate sacrificial material **102***b* directly formed on the substrate **101**.
(82) Thereafter, referring to operation S**170** of FIG. **11**, a portion of the first spacer material **180***a* may be removed through a stripping process. The first spacer material **180***a* may be removed, except for a portion thereof contacting the first epitaxial material EP**1***a* and the second epitaxial material EP**2***a* and a portion thereof formed in a first groove GV**1**. As the first spacer material **180***a* is partially removed, the first spacer material **180***a* may become first to third spacers **181** to **183** separate from one another.
(83) Subsequently, referring to operation S**180** of FIG. **12**, a process for growing the first epitaxial material EP**1***a* or the second epitaxial material EP**2***a* may be performed. For example, the first epitaxial material EP**1***a* may be grown such that the height of a top surface of the first epitaxial material EP**1***a* becomes equal to or higher than the heights of top surfaces of the substrate sacrificial material **102***b* and the substrate material **103***b*, which are disposed at uppermost sides. The grown first epitaxial material EP**1***a* and the second epitaxial material EP**2***a* may become a source/drain region **230**. The grown first epitaxial material EP**1***a* and the second epitaxial material EP**2***a* may become a first epitaxial region EP**1** and a second epitaxial region EP**2** of the source/drain region **230**, respectively.
(84) Although not clearly shown, a gate line **220** may then be formed in a region where the substrate sacrificial material **102***b*, the gate sacrificial pattern **162** and the hard mask pattern **163** are formed, through removal or replacement of the substrate sacrificial material **102***b*, the gate

sacrificial pattern **162** and the hard mask pattern **163**.
(85) Next, a semiconductor integrated circuit device according to example embodiments of the disclosure and a manufacturing method thereof will be described. In the following description, description will not necessarily be repeated of the same constituent elements as those of FIGS. **1** to **12**, and the same or similar reference numerals will be used for these constituent elements.
(86) FIG. **13** is a schematic cross-sectional view of a semiconductor integrated circuit device according to example embodiments of the disclosure.
(87) Referring to FIG. **13**, the semiconductor integrated circuit device according to some embodiments differs from the semiconductor integrated circuit device according to FIG. **2** in that a first section SP**1** and a second section SP**2** of a first spacer **181** include different materials, respectively.
(88) In some embodiments, the first section SP**1** and the second section SP**2** of the first spacer **181** may include different materials, respectively. The first section SP**1** and the second section SP**2** may be clearly distinguished from each other. For example, the first section SP**1** may include one of SiN, SiON or SiCON, and the second section SP**2** may include another one of SiN, SiON or SiCON. For example, the first section SP**1** and the second section SP**2** may be separate spacers including different materials, respectively.
(89) In some embodiments, the second section SP**2** of the first spacer **181** may include a material different from those of the first section SP**1** of the first spacer **181**, a second spacer **182** and a third spacer **183**. For example, the first section SP**1** of the first spacer **181**, the second spacer **182** and the third spacer **183** may include the same material.
(90) FIGS. **14** to **23** are views schematically showing procedures of a method for manufacturing the semiconductor integrated circuit device according to embodiments of FIG. **13**, respectively.
(91) Referring to operation S**210** of FIG. **14**, a substrate sacrificial material **102***a* and a substrate material **103***a* may be alternately stacked on a base substrate material **101**. The base substrate material **101** and the substrate material **103***a* may include the same material as the above-described substrate **101**. For example, the base substrate material **101** and the substrate material **103***a* may include Si. For example, the substrate sacrificial material **102***a* may include SiGe.
(92) Thereafter, an insulating pattern **161** may be formed on a structure in which the substrate sacrificial material **102***a* and the substrate material **103***a* are alternately stacked, and a plurality of vertical pattern structures, in which a gate sacrificial pattern **162** and a hard mask pattern **163** are stacked, may be formed on the insulating pattern **161**.
(93) Subsequently, an outer spacer material **158***a* covering or overlapping the plurality of vertical pattern structures may be formed. The outer spacer material **158***a* may include the same material as a fifth spacer **158**.
(94) Next, referring to operation S**220** of FIG. **15**, portions, among the plurality of vertical pattern structures, of the base substrate material **101** and the structure, in which the substrate sacrificial material **102***a* and the substrate material **103***a* are alternately stacked, may be removed through a wet etching process. The removed portions may have a predetermined depth. Vertical grooves may be formed by the removed portions. A side portion of the substrate sacrificial material **102***a* may be exposed at each vertical groove.
(95) A portion of an edge of the substrate sacrificial material **102***a* may be removed in each vertical groove through an over-etching process. A part of the exposed side portion of the substrate sacrificial material **102***a* may be removed. As the portion of the edge of the substrate sacrificial material **102***a* is removed, a horizontal stepped structure may be formed between a substrate sacrificial material **102***b* and a substrate material **103***b*. Two substrate materials **103***b* adjacent to each other and one substrate sacrificial material **102***b* formed therebetween may form a groove in accordance with a stepped structure thereof and, as such, first to third grooves GV**1** to GV**3** may be formed.
(96) Thereafter, referring to operation S**230** of FIG. **16**, a first spacer material **180***a* may be formed

to cover or overlap the vertical grooves while filling the first to third grooves GV**1** to GV**3** and to cover or overlap the plurality of vertical pattern structures. The first spacer material **180***a* may include the same material as the first section SP**1** of the first spacer **181**, the second spacer **182** and the third spacer **183**.
(97) Thereafter, referring to operation S**231** of FIG. **17**, a portion of the first spacer material **180***a* may be removed through a stripping process. The first spacer material **180***a* may be removed, except for a portion thereof contacting the first epitaxial material EP**1***a* and the second epitaxial material EP**2***a* and a portion thereof formed in the first groove GV**1**. As the first spacer material **180***a* is partially removed, the first spacer material **180***a* may become the first section SP**1** of the first spacer **181**, the second spacer **182** and the third spacer **183** separate from one another.
(98) Subsequently, referring to operation S**232** of FIG. **18**, a second spacer material **180***c* may be formed to cover or overlap the plurality of vertical pattern structures and the first spacer material **180***a*. The second spacer material **180***c* may include the same material as the second section SP**2** of the first spacer **181**.
(99) Next, referring to operation S**240** of FIG. **19**, a coating material **190** may be formed in a region where no process will be performed from among a first element region RX**1** and a second element region RX**2**. For example, the coating material **190** may be formed over the entire surface of the second element region RX**2**, for execution of a process in the first element region RX**1**. For example, the coating material **190** may include SOH.
(100) Thereafter, referring to operation S**250** of FIG. **20**, the coating material **190** may be removed through an etching process, and portions of the second spacer material **180***c* disposed on the hard mask pattern **163** and beneath the vertical grooves may be removed through an etch-back process. The partially-removed second spacer material **180***c* may expose a top surface of the substrate **101** and a top surface of the hard mask pattern **163**. The remaining portion is denoted as second spacer material **180***d*.
(101) Next, referring to operation S**260** of FIG. **21**, a first epitaxial material EP**1***a* and a second epitaxial material EP**2***a* may be sequentially formed on the substrate **101** exposed in the vertical grooves. The first epitaxial material EP**1***a* and the second epitaxial material EP**2***a* may be formed to have predetermined heights extending from the top surface of the substrate **101** while contacting the remaining portion of second spacer material **180***d*. In some embodiments, the height of a top surface of the second epitaxial material EP**2***a* may be equal to or lower than the height of a top surface of the substrate sacrificial material **102***b* directly formed on the substrate **101**.
(102) Thereafter, referring to operation S**270** of FIG. **22**, a portion of the second spacer material **180***c* may be removed through a stripping process. The second spacer material **180***c* may be removed, except for a portion thereof contacting the first epitaxial material EP**1***a* and the second epitaxial material EP**2***a* and a portion thereof contacting the first spacer material **180***a*. As the second spacer material **180***c* is partially removed, the second spacer material **180***c* may become the second section SP**2** of the first spacer **181**.
(103) Subsequently, referring to operation S**280** of FIG. **23**, a process for growing the first epitaxial material EP**1***a* or the second epitaxial material EP**2***a* may be performed. For example, the first epitaxial material EP**1***a* may be grown such that the height of a top surface of the first epitaxial material EP**1***a* becomes equal to or higher than the heights of top surfaces of the substrate sacrificial material **102***b* and the substrate material **103***b*, which are disposed at uppermost sides. The grown first epitaxial material EP**1***a* and the second epitaxial material EP**2***a* may become a source/drain region **230**. The grown first epitaxial material EP**1***a* and the second epitaxial material EP**2***a* may become a first epitaxial region EP**1** and a second epitaxial region EP**2** of the source/drain region **230**, respectively.
(104) Although not clearly shown, a gate line **220** may then be formed in a region where the substrate sacrificial material **102***b*, the gate sacrificial pattern **162** and the hard mask pattern **163** are formed, through removal or replacement of the substrate sacrificial material **102***b*, the gate

sacrificial pattern **162** and the hard mask pattern **163**.
(105) FIG. **24** is a schematic cross-sectional view of a semiconductor integrated circuit device according to example embodiments of the disclosure.
(106) Referring to FIG. **24**, the semiconductor integrated circuit device according to some embodiments differs from the semiconductor integrated circuit device according to the embodiments of FIG. **2** in that a plurality of channel active regions **111** and **112** is of a fin type or a planar type. For example, each channel active region may include a single channel. In addition, second to fourth channels CH**2** to CH**4** may be omitted. Furthermore, a portion of a first spacer **181**, a second spacer **182**, and a third spacer **183** may be omitted.
(107) In some embodiments, the first spacer **181** may be formed at opposite sides of the plurality of channel active regions **111** and **112**. The first spacer **181** may be formed on a first epitaxial region EP**1**. The first spacer **181** may overlap with the first epitaxial region EP**1** and a second epitaxial region EP**2** in a third direction D**3**, and may overlap with the channel active region in a first direction D**1**.
(108) In some embodiments, a gate line **220** may be formed on the plurality of channel active regions **111** and **112**.
(109) FIGS. **25** to **32** are views schematically showing procedures of a method for manufacturing the semiconductor integrated circuit device according to some embodiments of FIG. **24**, respectively.
(110) Referring to operation S**310** of FIG. **25**, an insulating pattern **161** may be formed on a base substrate material **101***a*, and a plurality of vertical pattern structures, in which a gate sacrificial pattern **162** and a hard mask pattern **163** are stacked, may be formed on the insulating pattern **161**. For example, the base substrate material **101***a* may include Si.
(111) Subsequently, an outer spacer material **158***a* covering or overlapping the plurality of vertical pattern structures may be formed. The outer spacer material **158***a* may include the same material as a fifth spacer **158**.
(112) Next, referring to operation S**320** of FIG. **26**, portions, among the plurality of vertical pattern structures, of the resultant base substrate material designated by reference numeral “**101**” may be removed through a wet etching process. Vertical grooves may be formed at the base substrate material **101** through removal of the portions of the base substrate material **101**. The vertical grooves may have a predetermined depth.
(113) Thereafter, referring to operation S**330** of FIG. **27**, a first spacer material **180***a* may be formed to cover or overlap the vertical grooves and to cover or overlap the plurality of vertical pattern structures. The first spacer material **180***a* may include the same material as the first spacer **181**.
(114) Next, referring to operation S**340** of FIG. **28**, a coating material **190** may be formed in a region where no process will be performed from among a first element region RX**1** and a second element region RX**2**. For example, the coating material **190** may be formed over the entire surface of the second element region RX**2**, for execution of a process in the first element region RX**1**. For example, the coating material **190** may include SOH.
(115) Thereafter, referring to operation S**350** of FIG. **29**, the coating material **190** may be removed through an etching process, and portions of the first spacer material **180***a* disposed on the hard mask pattern **163** and beneath the vertical grooves may be removed through an etch-back process. The partially-removed first spacer material **180***a* may expose a top surface of the base substrate material **101** becoming a substrate and a top surface of the hard mask pattern **163**. The remaining portion is denoted as first spacer material **180***b*.
(116) Next, referring to operation S**360** of FIG. **30**, a first epitaxial material EP**1***a* and a second epitaxial material EP**2***a* may be sequentially formed on the substrate **101** exposed in the vertical grooves. The first epitaxial material EP**1***a* and the second epitaxial material EP**2***a* may be formed to have predetermined heights extending from the top surface of the substrate **101** while contacting

the first spacer material **180***a*.
(117) Thereafter, referring to operation S**370** of FIG. **31**, a portion of the first spacer material **180***a* may be removed through a stripping process. The first spacer material **180***a* may be removed, except for a portion thereof contacting the first epitaxial material EP**1***a* and the second epitaxial material EP**2***a*. As the first spacer material **180***a* is partially removed, the first spacer material **180***a* may become the first spacers **181** separate from one another.
(118) Subsequently, referring to operation S**380** of FIG. **32**, a process for growing the first epitaxial material EP**1***a* or the second epitaxial material EP**2***a* may be performed. The grown first epitaxial material EP**1***a* and the second epitaxial material EP**2***a* may become a source/drain region **230**. The grown first epitaxial material EP**1***a* and the second epitaxial material EP**2***a* may become a first epitaxial region EP**1** and a second epitaxial region EP**2** of the source/drain region **230**, respectively.
(119) Although not clearly shown, a gate line **220** may then be formed in a region where the gate sacrificial pattern **162** and the hard mask pattern **163** are formed, through removal or replacement of the gate sacrificial pattern **162** and the hard mask pattern **163**.
(120) FIG. **33** is a schematic cross-sectional view of a semiconductor integrated circuit device according to example embodiments of the disclosure.
(121) Referring to FIG. **33**, the semiconductor integrated circuit device according to some embodiments differs from the semiconductor integrated circuit device according to the embodiments of FIG. **2** in that a first section SP**1** and a second section SP**2** of a first spacer **181** may be disposed to be spaced apart from each other.
(122) The first section SP**1** and the second section SP**2** of the first spacer **181** may include the same material or may include different materials, respectively. The first section SP**1** of the first spacer **181** may include the same material as a second spacer **182** and a third spacer **183**.
(123) In some embodiments, the first section SP**1** and the second section SP**2** of the first spacer **181** may be physically separated from each other. The first section SP**1** and the second section SP**2** of the first spacer **181** may include spacers that are separate from each other, respectively.
(124) One side surface of the second section SP**2** of the first spacer **181** may contact a first epitaxial region EP**1** and/or a second epitaxial region EP**2**, and the other side surface of the second section SP**2** of the first spacer **181** may contact a first channel CH**1**.
(125) FIG. **34** is a schematic cross-sectional view of a semiconductor integrated circuit device according to example embodiments of the disclosure.
(126) Referring to FIG. **34**, the semiconductor integrated circuit device according to some embodiments differs from the semiconductor integrated circuit device according to the embodiments of FIG. **2** in that a step is formed between top surfaces of a first section SP**1** and a second section SP**2** of a first spacer **181**.
(127) For example, the top surface of the second section SP**2** may be disposed between the top surface of the first section SP**1** and a bottom surface of the first section SP**1**.
(128) FIG. **35** is a schematic cross-sectional view of a semiconductor integrated circuit device according to example embodiments of the disclosure.
(129) Referring to FIG. **35**, the semiconductor integrated circuit device according to some embodiments differs from the semiconductor integrated circuit device according to the embodiments of FIG. **2** in that a first epitaxial region EP**1** is omitted from a source/drain region **230**.
(130) In some embodiments, the source/drain region **230** may be configured by a single epitaxial region. The source/drain region **230** may be of a type different from that of an element region where the source/drain region **230** is disposed.
(131) One side surface, a top surface and a bottom surface of a second section SP**2** of a first spacer **181** may contact the source/drain region **230**.
(132) FIG. **36** is a schematic cross-sectional view of a semiconductor integrated circuit device according to example embodiments of the disclosure.

(133) Referring to FIG. **36**, the semiconductor integrated circuit device according to some embodiments differs from the semiconductor integrated circuit device according to the embodiments of FIG. **2** in that a source/drain region **230** further includes a third epitaxial region EP**3**.
(134) The source/drain region **230** may include multiple epitaxial layers (for example, three layers or more). For example, the source/drain region **230** may further include the third epitaxial region EP**3** disposed between a first epitaxial region EP**1** and a second epitaxial region EP**2**. The third epitaxial region EP**3** may be of the same type as the first epitaxial region EP**1** or the second epitaxial region EP**2**. The third epitaxial region EP**3** may include a material different from that of the first epitaxial region EP**1** or the second epitaxial region EP**2**.
(135) In some embodiments, one side surface of the second section SP**2** of the first spacer **181** may contact the first epitaxial region EP**1** and the third epitaxial region EP**3**, a bottom surface of the second section SP**2** may contact the first epitaxial region EP**1**, and a top surface of the second section SP**2** may contact the second epitaxial region EP**2**.
(136) FIG. **37** is a schematic cross-sectional view of a semiconductor integrated circuit device according to example embodiments of the disclosure.
(137) Referring to FIG. **37**, the semiconductor integrated circuit device according to some embodiments differs from the semiconductor integrated circuit device according to the embodiments of FIG. **2** in that all portions of a second section SP**2** of a first spacer **181** are disposed to be higher than a first epitaxial region EP**1**.
(138) In some embodiments, one side surface of the second section SP**2** of the first spacer **181** may contact only the first epitaxial region EP**1**. In this case, a bottom surface of the second section SP**2** may contact a second epitaxial region EP**2**.
(139) FIG. **38** is a schematic cross-sectional view of a semiconductor integrated circuit device according to an example embodiment of the disclosure.
(140) Referring to FIG. **38**, the semiconductor integrated circuit device according to some embodiments differs from the semiconductor integrated circuit device according to the embodiments of FIG. **2** in that a first section SP**1** of a first spacer **181**, a second spacer **182** and a third spacer **183** are omitted.
(141) In some embodiments, the first spacer **181** may be formed only in a region penetrating a source/drain region **230**. The first spacer **181** may not contact a second channel CH**2**. The first spacer **181** may not overlap with the second channel CH**2** in a vertical/horizontal direction (for example, a first direction D**1** and a third direction D**3**).
(142) In accordance with the example embodiments of the disclosure, it may be possible to minimize or reduce leakage of current through a lower side of a channel between adjacent source/drain regions.
(143) While the embodiments of the disclosure have been described with reference to the accompanying drawings, it should be understood by those skilled in the art that various modifications may be made without departing from the scope of the disclosure. Therefore, the above-described embodiments should be considered in a descriptive sense only and not for purposes of limitation.

Claims

1. A semiconductor integrated circuit device comprising: a substrate comprising a first element region of a P type and a second element region of an N type; a channel active region that extends into the first element region or the second element region in a first direction, wherein the channel active region comprises a plurality of channels; a plurality of gate lines that extend in a second direction intersecting the first direction, wherein each of the plurality of gate lines comprises a gate metal layer and a gate insulating film in contact with the gate metal layer; a plurality of first spacers

on opposite side portions of the plurality of gate lines, wherein the plurality of first spacers are separated from one another; and a plurality of source/drain regions that are between the plurality of gate lines, wherein the channel active region comprises a first channel directly on the substrate, and a second channel spaced apart from the first channel in a third direction perpendicular to the first direction and the second direction and extends into the gate metal layer, wherein a width in the first direction of the second channel is greater than respective widths in the first direction of the gate metal layer and the gate insulating film that are between the first channel and the second channel, wherein the first channel, the second channel, and portions of the gate lines between the first channel and the second channel have a first groove therebetween, wherein a first spacer of the plurality of first spacers comprises a first section in the first groove and a second section that penetrates a first source/drain region of the plurality of source/drain regions, and wherein a lower insulating film is disposed directly on the substrate on opposite side portions of the first channel in the second direction.
2. The semiconductor integrated circuit device according to claim 1, wherein the first section of the first spacer overlaps the second channel in the third direction, and wherein the second section of the first spacer overlaps the first source/drain region in the third direction.
3. The semiconductor integrated circuit device according to claim 2, wherein a thickness of the second section of the first spacer is greater than a thickness of the first section of the first spacer.
4. The semiconductor integrated circuit device according to claim 2, wherein the first section of the first spacer contacts the gate insulating film.
5. The semiconductor integrated circuit device according to claim 1, wherein the first source/drain region of the plurality of source/drain regions comprises: a first epitaxial region on the substrate; and a second epitaxial region on the first epitaxial region and having a second conductivity type different from a first conductivity type of the first epitaxial region.
6. The semiconductor integrated circuit device according to claim 5, wherein the second channel contacts respective second epitaxial regions of ones of the plurality of source/drain regions at opposite side portions of the second channel.
7. The semiconductor integrated circuit device according to claim 5, wherein the second section of the first spacer is on the first epitaxial region.
8. The semiconductor integrated circuit device according to claim 5, wherein a top surface of the first epitaxial region is lower than a bottom surface of the second channel with respect to the substrate.
9. The semiconductor integrated circuit device according to claim 5, wherein the first epitaxial region is of the P type and the channel active region is in the first element region, and wherein the first epitaxial region is of the N type and the channel active region is in the second element region.
10. The semiconductor integrated circuit device according to claim 5, wherein the channel active region further comprises a third channel spaced apart from the second channel in the third direction and extending through the gate metal layer, and wherein the third channel contacts respective second epitaxial regions of ones of the source/drain regions at opposite sides of the third channel.
11. The semiconductor integrated circuit device according to claim 10, wherein the second channel, the third channel, and portions of the gate lines between the second channel and the third channel have a second groove therebetween.
12. The semiconductor integrated circuit device according to claim 11, further comprising: a plurality of second spacers on the opposite side portions of the gate lines in the second groove.
13. The semiconductor integrated circuit device according to claim 12, wherein the plurality of second spacers do not overlap with any of the plurality of source/drain regions in the third direction.
14. The semiconductor integrated circuit device according to claim 1, wherein the first section of the first spacer and the second section of the first spacer comprise different materials, respectively, and wherein the first section of the first spacer and the second section of the first spacer are

separated from each other.
15. The semiconductor integrated circuit device according to claim 1, wherein the gate insulating film comprises a ferroelectric material film having ferroelectric characteristics and a negative capacitance, and a paraelectric material film having paraelectric characteristics and a positive capacitance.
16. A semiconductor integrated circuit device comprising: a substrate comprising a first element region of a P type and a second element region of an N type; a channel active region that extends into the first element region or the second element region in a first direction, wherein the channel active region comprises a plurality of channels; a plurality of gate lines that extend in a second direction intersecting the first direction, wherein each of the plurality of gate lines comprises a gate metal layer and a gate insulating film in contact with the gate metal layer; source/drain regions that are between the plurality of gate lines, wherein a first source/drain region of the source/drain regions comprises a first epitaxial region and a second epitaxial region of different conductivity types; and a plurality of first spacers on opposite side portions of the plurality of gate lines, wherein a first spacer of the plurality of first spacers is between the first epitaxial region and the second epitaxial region of the first source/drain region, and overlaps a respective one of the plurality of gate lines in the first direction, and wherein a lower insulating film is disposed directly on the substrate on opposite side portions of one of the plurality of channels in the second direction.
17. The semiconductor integrated circuit device according to claim 16, wherein the first spacer comprises: a first section that contacts the gate insulating film; and a second section that contacts both the first epitaxial region and the second epitaxial region of the first source/drain region.
18. The semiconductor integrated circuit device according to claim 17, wherein a top surface of the second section of the first spacer contacts the first epitaxial region, and a bottom surface of the second section of the first spacer contacts the second epitaxial region.
19. The semiconductor integrated circuit device according to claim 16, wherein the first epitaxial region is of a same conductivity type as the first element region or the second element region into which the channel active region extends.
20. A semiconductor integrated circuit device comprising: a substrate comprising a first element region of a P type and a second element region of an N type; a channel active region that extends into the first element region or the second element region in a first direction, wherein the channel active region comprises a plurality of channels; a plurality of gate lines that extend in a second direction intersecting the first direction, wherein each of the plurality of gate lines comprises a gate metal layer and a gate insulating film in contact with the gate metal layer; a plurality of spacers on opposite side portions of the gate lines, wherein the plurality of spacers are separated from one another; and a plurality of source/drain regions that are between the plurality of gate lines, wherein the channel active region comprises: a first channel directly on the substrate, a second channel spaced apart from the first channel in a third direction perpendicular to the first direction and the second direction and extends into the gate metal layer, and a third channel spaced apart from the second channel in the third direction and extending into the gate metal layer, wherein respective widths in the first direction of the second channel and the third channel are greater than respective widths in the first direction of the gate metal layer and the gate insulating film that are between the first channel and the second channel, wherein the first channel, the second channel, and respective portions of first ones of the gate lines between the first channel and the second channel have a first groove therebetween, wherein the second channel, the third channel, and respective portions of second ones of the gate lines between the second channel and the third channel have a second groove therebetween, wherein the plurality of spacers comprises: a first spacer comprising a first section in the first groove, and a second section that penetrates a first source/drain region of the plurality of source/drain regions, a second spacer in the second groove, a third spacer on the third channel at an outside of a part of a respective one of the gate lines, and a fourth spacer on the third channel at an outside of the third spacer, wherein the first source/drain region comprises a first

epitaxial region and a second epitaxial region of different conductivity types, wherein the first spacer is directly on the first epitaxial region and is directly on the second epitaxial region, wherein the second spacer contacts the second epitaxial region, and wherein a lower insulating film is disposed directly on the substrate on opposite side portions of the first channel in the second direction.